Pierre Emery

514-441-9866 | Montreal | pierre.emery@mila.quebec | github.com/pierre-emery | linkedin.com/in/pierre-emery-

Education

Mila - Quebec's AI Institute / Université de Montréal Sept 2026 — May 2028
Professional M.Sc. in Machine Learning *Montreal, QC*

- Coursework: Probabilistic Graphical Models, Representation Learning

Université de Montréal Sept 2023 — Aug 2026
B.Sc. in Mathematics and Computer Science - Data Science Profile *Montreal, QC*

- Coursework: Statistical Learning, Bayesian Statistics, Theoretical Foundations of Data Science
- Exchange semester at **EPFL, Lausanne** (Spring 2025)

Experience

Software Development Intern May 2026 — Aug 2026
Centre d'expertise et de recherche en infrastructures urbaines (CERIU) *Montreal, QC*

Python, FastAPI, Pydantic, Power BI (DAX, Power Query M), Tkinter, Microsoft Graph

- Extended an inherited two-source prototype into a FastAPI backend deployed on Render, integrating five external APIs (GA4, Mailchimp, Vimeo, Facebook, LinkedIn); packaged the connectors as CanalPy, a pip-installable library published on TestPyPI.
- Diagnosed that per-item API calls (Facebook post insights, LinkedIn posts, Mailchimp click reports) exceeded Power BI's refresh window even when parallelized with `ThreadPoolExecutor`; offloaded them to a standalone Tkinter tool writing CSVs to SharePoint via Microsoft Graph.
- Designed for handoff to a non-technical team: moved credentials from server-side environment variables to Power BI parameters, refactored 50+ Power Query queries around reusable M functions, and wrote an MkDocs maintenance guide.

Projects

Glacier Segmentation from Satellite Imagery (github.com/pierre-emery/projet_glacier) Jan 2026 — Apr 2026

Python, PyTorch, GeoPandas, rioxarray, STAC API

- Built the full data pipeline from raw Sentinel-2 L2A scenes to 582 curated image–mask pairs: SCL-based cloud/shadow/snow quality score, median compositing of the best late-summer scenes, and GLIMS outlines rasterized into aligned masks.
- Trained an Attention U-Net (≈7.8M params) on Caucasus glaciers only; it reached 0.83 test IoU on unseen mountain ranges (0.82 Alps, 0.84 Andes). An ablation showed augmentation drives most of the gain, with dropout and weight decay helping only in combination with it.
- Adapted an ImageNet-pretrained DeepLabV3+ (≈42M params) to 5-band input by initializing NIR/SWIR filters from the mean RGB weights; the U-Net outperformed it (0.83 vs. 0.81 IoU) with 5× fewer parameters.

NHL Expected Goals (xG) Modeling (github.com/Stephanope/Project-Hockey) Dec 2025 — Feb 2026

Python, pandas, scikit-learn, XGBoost, Weights & Biases

- Graduate-level project (IFT6758) taken on with a friend: built a cached play-by-play pipeline over the NHL API and engineered 19 shot-level features, including coordinates standardized via inferred attack side, rebounds, pre-shot speed and power-play state.
- Showed that 90.6% baseline accuracy only reflected class imbalance and used ranking and calibration metrics instead; an XGBoost model tuned on log loss raised AUC from 0.72 to 0.78 vs. logistic regression, with SHAP analysis in a blog post.

Skills

- **Programming**: Python, SQL, R
- **Libraries**: PyTorch, scikit-learn, XGBoost, SHAP, NumPy, pandas, SciPy, Plotly, GeoPandas, rioxarray
- **Tools**: Git, Docker, FastAPI, Pydantic, Google Cloud, Weights & Biases, STAC API, Power BI (DAX, Power Query M)

Languages and Interests

- **Languages**: French (native), English (native)
- **Interests**: Bouldering, mountaineering & wildlife photography